MODULE 1 — OVERVIEW OF COMPUTER-GENERATED WORLDS

1. Introduction to Virtual and Augmented Reality: Computer-Generated Worlds

What is Virtual Reality?

Virtual Reality (VR) is a technology that creates a computer-generated three-dimensional environment and gives the user the feeling of being present inside that environment.

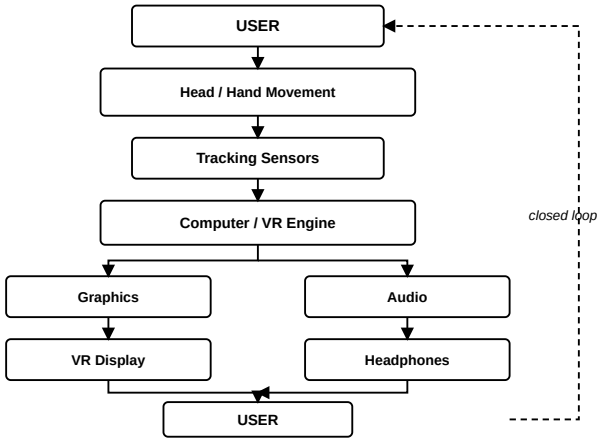
Instead of looking at a virtual world through a normal monitor, the user generally uses a Head-Mounted Display (HMD). The system also tracks the user's movements and changes the virtual environment accordingly.

For example, suppose you are inside a VR flight simulator. You look to your left, and you see the left side of the aircraft cockpit. You look upward, and the ceiling appears. You move your head down, and you see the controls.

The computer is continuously changing the displayed scene according to your movement.

Basic VR interaction

DIAGRAM



Basic VR interaction loop

This forms a closed interaction loop.

The user performs an action → the system detects it → the computer updates the virtual world → the user receives new visual/audio/haptic information.

What is a computer-generated world?

A computer-generated world is an artificial environment created using computer software.

It can contain:

- 3D objects
- Buildings
- Characters
- Landscapes
- Lighting
- Shadows
- Textures
- Sound
- Animation
- Physical behaviour

For example, in a VR driving simulator, the computer can generate:

```
Road + Cars + Buildings + Trees + Traffic + Weather
```

The user does not need to physically travel through the environment. The computer creates a digital representation of it.

The process of converting the virtual scene into an image that can be displayed is called rendering.

What is Augmented Reality?

Augmented Reality (AR) adds computer-generated information or virtual objects to the user's view of the real world.

Unlike VR, AR does not completely replace reality.

For example, imagine pointing your phone camera at your room and seeing a virtual sofa placed on the floor.

The room is real.

The sofa is virtual.

Therefore:

```
AR = Real Environment + Computer-Generated Content
```

Examples include:

- Pokémon GO
- AR navigation
- Furniture placement applications
- Industrial maintenance systems
- AR glasses

VR vs AR

VR	AR
Replaces the user's environment with a virtual environment	Adds virtual information to the real environment
High immersion	Usually lower immersion
Uses VR headset/HMD	Uses phone, AR glasses or AR headset
User is mainly inside the virtual world	User remains aware of the real world
Example: flight simulator	Example: AR navigation

Diagram required: YES

Learn the VR interaction-loop diagram and a simple comparison:

```
VR:
REAL WORLD → VIRTUAL WORLD

AR:
REAL WORLD → VIRTUAL CONTENT
```

2. History of Virtual and Augmented Reality

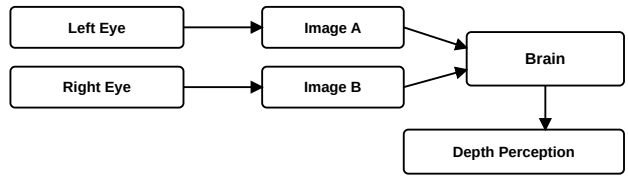
The development of VR and AR was gradual. It evolved from simple stereoscopic images into highly interactive computer-generated environments.

Early stereoscopic systems

One of the fundamental principles behind VR is stereoscopic vision.

Our two eyes see slightly different images of the same scene. The brain combines these images to perceive depth.

DIAGRAM



Stereoscopic vision

This principle was used in early stereoscopic viewing devices.

Sensorama — 1950s

Morton Heilig developed the Sensorama in the 1950s.

It attempted to provide a multisensory experience using:

- Visual stimulation
- Sound
- Smell
- Motion

The important idea was that a simulated experience should involve multiple human senses, not just vision.

This concept later became fundamental to immersive VR.

Sword of Damocles — 1968

In 1968, Ivan Sutherland and Bob Sproull developed an early computer-connected head-mounted display known as the Sword of Damocles.

It could display simple computer-generated graphics.

It was extremely large and was mechanically suspended above the user's head.

It is considered one of the important foundations of modern VR and AR.

Development of computer graphics

During the 1970s and 1980s, computer graphics improved considerably.

Computers became capable of generating:

- 2D graphics
- 3D objects
- Polygons
- Textures
- Animation
- Simulated environments

This was essential for VR because a VR system needs to generate and update the virtual environment in real time.

Development of interactive VR

During the 1980s, Jaron Lanier and VPL Research contributed significantly to the development and popularization of VR.

Systems such as:

- Head-mounted displays
- DataGlove
- Body tracking

allowed users to interact with virtual environments.

This was a major change.

VR was no longer simply:

"Look at a computer-generated image."

It became:

"Enter a computer-generated environment and interact with it."

Development of AR

AR developed from the idea of combining computer-generated information with the real environment.

Research in the late 20th century produced systems that could display digital information in relation to physical environments.

A notable example was Virtual Fixtures, developed by Louis Rosenberg in the early 1990s.

The system demonstrated how virtual information could assist a user performing physical tasks.

Modern VR and AR

The development of:

- GPUs
- High-resolution displays
- Cameras
- Motion sensors
- Accelerometers
- Gyroscopes
- Computer vision
- Smartphones
- Wireless communication

made VR and AR much more practical.

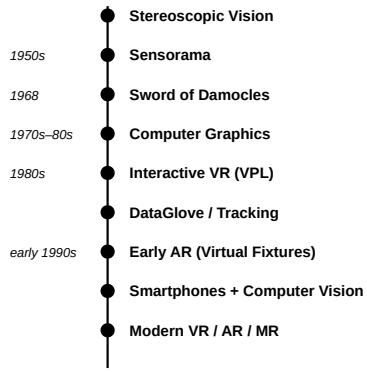
Modern systems can track:

- Head position
- Head orientation
- Hand movement
- Body movement
- Physical surfaces
- Real-world objects

This has produced modern consumer VR, AR and Mixed Reality systems.

Historical development

DIAGRAM



Historical development of VR and AR

Diagram required: YES

For this question, draw a timeline. It is much better than writing only dates.

3. The Virtuality Continuum

The Virtuality Continuum explains the range between the completely real environment and the completely virtual environment.

It was proposed by Paul Milgram and Fumio Kishino in their work on mixed reality.

The key idea is that reality and virtuality are not necessarily two separate categories.

There can be many combinations between them.

Real Environment

At one extreme is the real world.

For example:

You are sitting in your classroom and looking at your actual surroundings.

Everything you see is physically present.

Augmented Reality

Move toward the virtual side.

The real world remains visible, but virtual content is added.

```
REAL WORLD
+
VIRTUAL OBJECTS
↓
AUGMENTED REALITY
```

Example:

You look at a real engine through AR glasses and see:

```
Temperature = 82°C
```

displayed next to the engine.

Augmented Virtuality

Now move further toward the virtual side.

The environment is primarily virtual, but real-world information is incorporated into it.

For example, a virtual machine model could receive live temperature information from a physical machine.

```
VIRTUAL ENVIRONMENT
+
REAL-WORLD DATA
↓
AUGMENTED VIRTUALITY
```

Virtual Environment

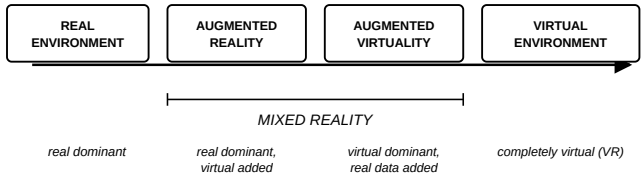
At the opposite extreme is a completely virtual environment.

The environment is primarily computer-generated.

This corresponds to VR.

Continuum

DIAGRAM



The Virtuality Continuum

The most important thing to understand is:

AR: Real environment is dominant, virtual information is added.

AV: Virtual environment is dominant, real information is added.

Practical example

Consider engineering.

Real environment

Engineer physically examines an engine.

AR

Engineer looks at the real engine while digital temperature, pressure and maintenance instructions are displayed.

Augmented Virtuality

Engineer enters a virtual representation of the engine, but live data from the physical engine is incorporated into the virtual model.

VR

Engineer examines a completely simulated virtual engine.

Diagram required: YES — VERY IMPORTANT

You should definitely learn the Real → AR → AV → Virtual continuum diagram.

4. Architecture of Virtual and Augmented Reality

This topic is about how a VR/AR system is organized internally and how its major components communicate.

Don't think of architecture as a building. Here, architecture means the structure of the system.

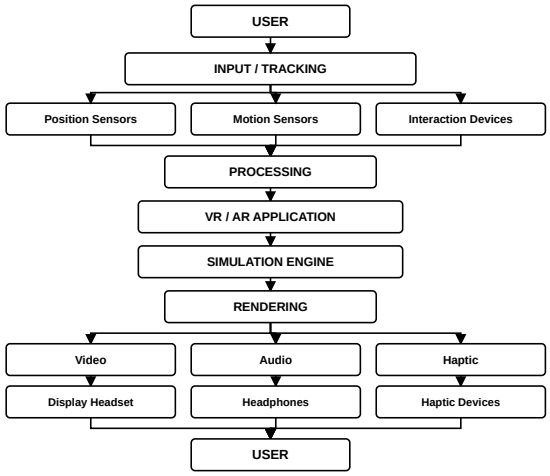
A VR/AR system generally contains:

1. Input and tracking
2. Processing
3. Simulation/application

4. Rendering
5. Output/display
6. Interaction and feedback

Basic architecture

DIAGRAM



Basic VR/AR architecture

Input and tracking

The system first needs to know what the user is doing.

Sensors can detect:

- Head position
- Head orientation
- Hand position
- Controller movement
- Body movement
- Eye movement

Common sensors include:

- Accelerometers
- Gyroscopes
- Cameras
- Magnetometers
- Optical trackers

For example:

If you turn your head right:

```
Head rotates
↓
Gyroscope / camera detects movement
↓
System calculates new orientation
↓
Virtual camera changes
```

Processing

The computer processes sensor information and determines what should happen in the virtual environment.

For example:

```
User moves hand
↓
Tracking data
↓
Computer calculates hand position
↓
Virtual hand moves
```

Application and simulation

This is where the actual virtual world is controlled.

It contains information about:

- Objects
- Positions
- Physics
- Rules
- Interactions
- Environment
- User actions

For example, if you throw a virtual ball, the simulation calculates its trajectory.

Rendering

Rendering converts the virtual scene into images that can be displayed.

The rendering system considers:

- Camera position
- Object position
- Lighting
- Texture
- Depth
- Perspective

VR often needs separate images for the left and right eyes to create stereoscopic depth.

Output

The system provides information to the user through multiple sensory channels.

Visual

VR/AR displays.

Audio

Headphones or spatial audio systems.

Haptic

Vibration and force feedback devices.

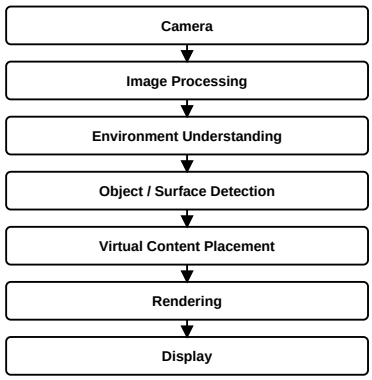
AR-specific architecture

AR has an additional important requirement:

The system must understand the real environment.

Therefore, AR commonly uses:

DIAGRAM



AR-specific architecture

For example, an AR application may detect the floor before placing a virtual chair on it.

Diagram required: YES — VERY IMPORTANT

Learn the complete VR/AR architecture block diagram.

5. Applications of Virtual and Augmented Reality — Gaming and Entertainment

Gaming is one of the most visible applications of VR and AR.

Traditional games provide interaction through:

- Keyboard
- Mouse
- Controller
- Screen

VR changes this by placing the player inside the game environment.

VR gaming

In VR gaming, the player wears an HMD.

The system tracks the player's:

- Head
- Hands
- Controllers
- Sometimes body

The virtual environment changes according to the player's movement.

For example:

```
Player turns head
↓
Head tracking
↓
Virtual camera rotates
↓
Game scene changes
```

This creates a stronger feeling of presence.

Immersion

VR games can provide a high level of immersion.

For example, in a VR horror game, the player can actually look around a virtual room rather than simply viewing it on a screen.

The system can use:

- 3D graphics
- Spatial audio

- Haptic feedback
- Motion tracking

to increase realism.

AR gaming

AR gaming combines game elements with the physical world.

A famous example is Pokémon GO.

The player sees:

```
Real street
↓
Virtual Pokémon
↓
AR Game
```

The real environment becomes part of the gameplay.

Entertainment applications

VR and AR are also used in:

Virtual cinema

Users can experience a virtual theater environment.

Virtual concerts

Users can attend digitally generated or immersive concert environments.

Sports

AR can provide:

- Player statistics
- Score information
- Visual overlays

Theme parks

VR can be combined with physical rides to create simulated environments.

Virtual tourism

Users can explore:

- Museums
- Historical locations
- Tourist destinations

without physically being there.

Advantages

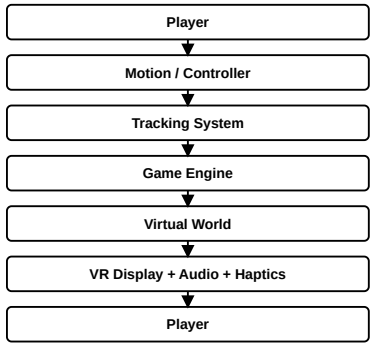
VR/AR can provide:

- Immersive experiences
- Interactive gameplay
- Personalized experiences
- Realistic simulation
- New forms of storytelling

Diagram required: YES

A simple gaming interaction diagram is enough:

DIAGRAM



Gaming interaction diagram

6. Applications of Virtual and Augmented Reality — Science and Engineering

VR and AR are highly useful in science and engineering because they allow users to visualize complex structures and systems.

Many scientific objects are difficult to see directly.

For example:

- Molecules
- Human organs
- Internal structures
- Aircraft
- Machines
- Buildings
- Bridges

VR/AR can represent these systems in 3D.

Scientific visualization

Scientists can create virtual models of complex systems.

For example, a molecule can be represented as a 3D structure.

Instead of looking at:

```
2D image
```

the researcher can interact with:

```
3D molecular model
```

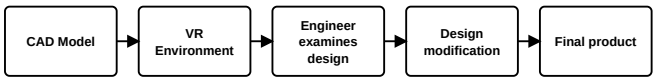
They can rotate, zoom and inspect it.

Engineering design

Engineers can create virtual prototypes before manufacturing the physical product.

For example:

DIAGRAM



Engineering design workflow

This can reduce the need for multiple physical prototypes.

Architecture

Architects can create virtual walkthroughs.

Before construction, a person can enter a virtual building and inspect:

- Room size
- Furniture placement
- Lighting
- Doors
- Windows
- Interior design

Maintenance

AR can assist engineers in maintaining machines.

An engineer looking at a machine through AR glasses could see:

```
Motor
↓
Temperature: 75°C

Valve
↓
Maintenance Required

Panel
↓
Open Here
```

This provides information directly in the engineer's field of view.

Training

VR can simulate dangerous engineering situations without exposing people to actual danger.

Examples:

- Aircraft maintenance
- Nuclear plant training
- Chemical plant operation
- Heavy machinery operation

Advantages

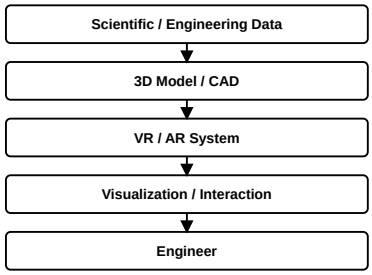
VR/AR in science and engineering provides:

- 3D visualization
- Better understanding
- Virtual prototyping
- Safer training
- Improved maintenance
- Reduced prototype costs
- Better collaboration

Diagram required: YES

Use:

DIAGRAM



Science and engineering workflow

7. Applications of Virtual and Augmented Reality — Medicine

Medicine is one of the most important applications of VR and AR.

The technologies can be used for:

- Medical training
- Surgical planning
- Surgical assistance
- Rehabilitation
- Therapy
- Medical visualization
- Patient education

Medical training

Medical students can use VR to study anatomy.

Instead of studying only from a textbook, students can examine a 3D virtual human body.

They can:

- Rotate organs
- Zoom in
- Remove layers
- Examine internal structures
- Study relationships between organs

For example:

```
Virtual Human Body
↓
Select organ
↓
Isolate organ
↓
Rotate / Zoom
↓
Study anatomy
```

Surgical simulation

VR can simulate surgical procedures.

A student can practice:

```
Incision
↓
Tissue manipulation
↓
Instrument movement
↓
Surgical procedure
```

without operating on a real patient.

This provides a safe training environment.

Surgical planning

Doctors can create 3D models from medical imaging data such as CT or MRI.

For example:

```
CT / MRI Data
↓
3D Reconstruction
↓
Virtual Organ Model
↓
Doctor examines anatomy
↓
Surgical planning
```

This can help doctors understand complicated anatomy before surgery.

Augmented Reality in surgery

AR can display useful information while the doctor works with a real patient.

For example, an AR system could display:

- Anatomical structures
- Surgical guidance
- Patient information
- Instrument positioning

The physical patient remains visible while digital information is overlaid.

Rehabilitation

VR can create controlled environments for rehabilitation exercises.

For example, a patient recovering from an injury may perform virtual tasks that require controlled movement.

The system can track movement and measure progress.

Therapy

VR has also been explored for therapeutic applications such as controlled exposure environments.

For example, a controlled virtual environment can be created for exposure-based therapeutic exercises under professional supervision.

Advantages

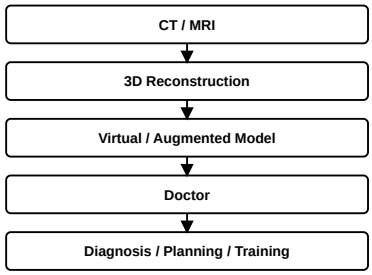
VR/AR can provide:

- Safe medical training
- Better visualization
- Surgical planning assistance
- Interactive anatomy education
- Rehabilitation environments
- Reduced training risk

Diagram required: YES

Learn:

DIAGRAM



Medical workflow

8. Applications of Virtual and Augmented Reality — Education

VR and AR can change education from simply reading or watching into experiencing and interacting.

VR in education

VR can create an immersive classroom or laboratory.

For example, instead of reading about ancient Egypt, students can enter a virtual reconstruction of an ancient Egyptian city.

They can look around and interact with the environment.

Virtual laboratories

Students can perform experiments inside a virtual laboratory.

For example:

```
Virtual Laboratory
↓
Select equipment
↓
Perform experiment
↓
Observe result
↓
Repeat experiment
```

This is especially useful when real equipment is:

- Expensive
- Dangerous
- Unavailable
- Difficult to access

History

Students can experience historical environments.

For example:

- Ancient civilizations
- Historical buildings
- Archaeological sites
- Historical events

Instead of simply reading:

"This building was constructed in the 15th century."

the student can explore a virtual reconstruction.

Science

VR can visualize concepts that are difficult to observe directly.

For example:

- Solar system
- Human anatomy

- Molecular structures
- Chemical reactions
- Geological structures

AR in education

AR can overlay information on physical educational materials.

For example, a textbook could contain an image of the human heart.

Using an AR application:

```
Textbook
+
Camera
↓
Heart recognized
↓
3D heart appears
↓
Student interacts with it
```

Benefits

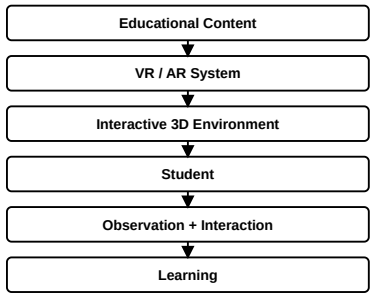
VR and AR can provide:

- Interactive learning
- Better visualization
- Increased engagement
- Practical training
- Safe experimentation
- Experiential learning
- Remote access to simulated environments

However, there can also be limitations such as:

- Hardware cost
- Technical requirements
- Motion sickness in some VR users
- Need for suitable educational content
- Teacher training

DIAGRAM



Educational workflow

9. Telerobotics and Telepresence

This is a particularly important topic because it combines robotics, networking, sensors, VR and human interaction.

Let's separate the two concepts first.

Telerobotics

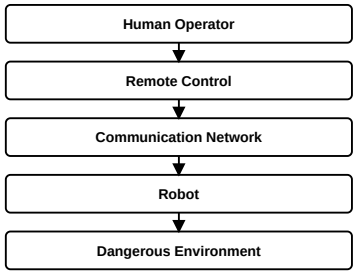
Telerobotics is the use of a human operator to control a robot remotely.

The operator may be hundreds or thousands of kilometres away from the robot.

For example, imagine a robot working in a dangerous chemical environment.

Instead of sending a human into the area:

DIAGRAM



Telerobotics

The human controls the robot from a safe location.

Why telerobotics is useful

It is useful when the environment is:

- Dangerous
- Inaccessible
- Remote
- Too small
- Too hot
- Radioactive
- Underwater

Examples include:

- Space exploration
- Underwater robots
- Nuclear plant inspection
- Bomb disposal
- Remote industrial operations

Telepresence

Telepresence is the feeling of being present in a remote environment through technology.

The difference is important.

Telerobotics focuses mainly on:

Controlling a remote robot.

Telepresence focuses on:

Giving the user the perception of being present at the remote location.

VR plays an important role here.

How telepresence works

Suppose a robot is located in a remote environment.

The robot contains:

- Cameras
- Microphones
- Sensors
- Motion sensors

The data is transmitted to the human operator.

The operator may wear:

- VR headset
- Headphones
- Haptic gloves

Now the operator can see and hear the remote environment.

If the operator turns their head:

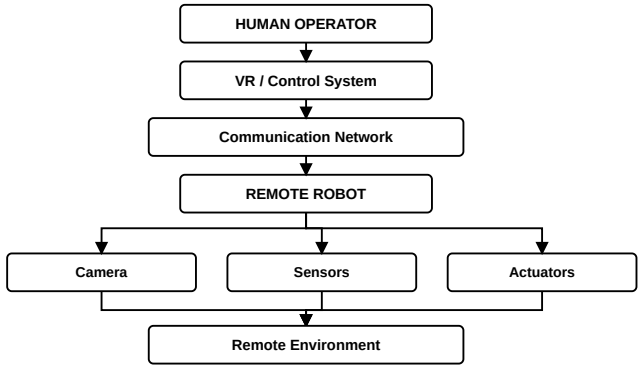
```
Operator turns head
↓
Head tracking
↓
Command sent to remote robot
↓
Robot camera changes direction
↓
New video sent back
↓
Operator sees remote environment
```

This creates the feeling of being physically present at the remote location.

Telerobotics + Telepresence

These technologies can work together.

DIAGRAM



The remote robot sends sensory information back to the operator; the operator sends commands to the robot.

Telerobotics + Telepresence

The remote robot sends sensory information back to the operator.

The operator sends commands to the robot.

Therefore, communication is bidirectional.

```
Operator ——— Commands ———→ Robot
Operator ←——— Sensor Data ——— Robot
```

Example — Space exploration

Imagine a robot on Mars.

A human operator on Earth cannot physically travel to Mars just to control it directly.

Instead:

```
Earth
Human Operator
↓
Control Commands
↓
Communication Link
↓
Mars Rover
↓
Camera + Sensors
↓
Mars Environment
```

The rover collects:

- Images
- Video
- Environmental measurements
- Other sensor information

and sends them back.

The operator uses these data to understand and control the remote environment.

Example — Remote surgery

A related application is remote robotic surgery.

The surgeon controls robotic instruments from another location.

Conceptually:

DIAGRAM



Remote surgery

The system can provide visual and sometimes haptic feedback to the surgeon.

This requires extremely reliable communication and precise control.

Difference between Telerobotics and Telepresence

Telerobotics	Telepresence
Remote control of robots	Feeling of being present remotely
Main focus is robot operation	Main focus is human perception/presence
Uses commands and actuators	Uses video, audio, sensors and immersive interfaces
Example: controlling Mars rover	Example: seeing/hearing through a remote robot
VR may or may not be used	VR can strongly enhance the experience

Advantages

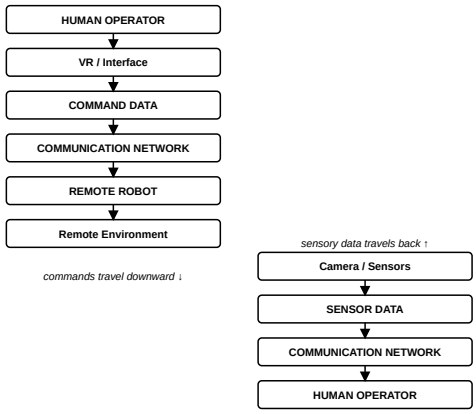
Telerobotics and telepresence allow humans to:

- Work in dangerous environments
- Operate remote machines
- Explore inaccessible locations
- Reduce physical risk
- Perform remote inspection
- Conduct remote operations
- Explore space and deep-sea environments

Diagram required: YES — VERY IMPORTANT

Learn the bidirectional communication diagram:

DIAGRAM

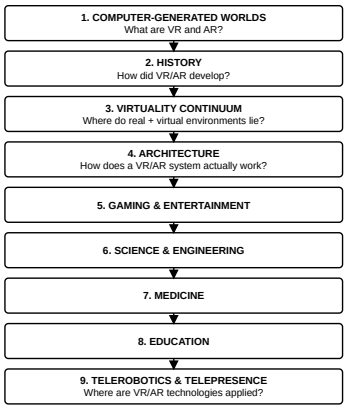


Bidirectional communication in telerobotics

How the 9 topics fit together

Don't memorize these as nine unrelated answers. There is a logical progression through the module:

DIAGRAM



How the 9 topics fit together

MODULE 2 — RELATIONSHIP TO OUTPUT AND INPUT DEVICES

1. Mechanics of Sight

What does "Mechanics of Sight" mean?

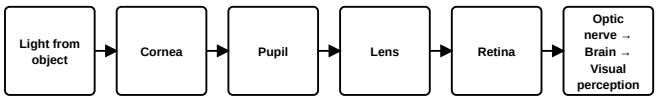
The mechanics of sight explains how the human visual system receives light from the environment, forms images in the eyes, and allows the brain to perceive objects, depth, motion and spatial information.

In VR/AR, understanding the mechanics of sight is important because displays must reproduce visual information in a way that is compatible with human vision.

How human sight works

The basic process is:

DIAGRAM



How human sight works

The cornea and lens focus incoming light onto the retina.

The retina contains photoreceptor cells that respond to light. The resulting signals travel through the optic nerve to the brain, where visual information is interpreted.

Important aspects for VR

Binocular vision

Humans have two eyes separated by a small distance.

Each eye receives a slightly different image.

The brain combines these images to perceive depth.

```
Left eye  → Image A
Right eye → Image B
        ↓
      Brain
        ↓
Depth perception
```

This is why VR displays provide slightly different images to each eye.

Field of View

The Field of View (FOV) is the extent of the environment visible to the user.

A wider FOV generally provides a more immersive visual experience.

Focus and accommodation

The eye changes the shape of its lens to focus objects at different distances. This process is called accommodation.

VR systems must consider how displayed depth relates to the eye's focusing behaviour.

3-mark answer structure

Mechanics of sight refers to the process by which the human visual system receives light and converts it into visual perception. Light enters through the cornea and pupil, is focused by the lens onto the retina, and signals are transmitted through the optic nerve to the brain. Binocular vision, field of view and depth perception are particularly important in VR because VR displays provide separate images to the two eyes to create a 3D visual experience.

Diagram: Recommended

Learn the simple:

Object → Cornea → Pupil → Lens → Retina → Brain

2. Component Technologies of Head-Mounted Displays

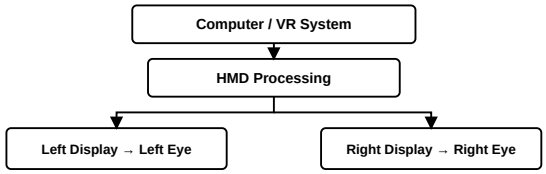
What is a Head-Mounted Display?

A Head-Mounted Display (HMD) is a display device worn on the user's head that presents visual information directly in front of the user's eyes.

A VR headset such as a Meta Quest is an HMD.

The basic idea is:

DIAGRAM



Basic idea of an HMD

The HMD doesn't simply show a normal screen.

It has to create the illusion of a three-dimensional environment.

To achieve this, several technologies work together.

Main components of an HMD

The important components are:

1. Display system
2. Optical system
3. Stereo vision system
4. Tracking system
5. Audio system
6. Headset enclosure
7. Processing system
8. Adjustment mechanisms

Let's understand each.

1. Display system

The display is the component that generates the visual scene.

Modern HMDs commonly use technologies such as:

- OLED
- AMOLED
- LCD
- Micro-OLED

The display needs to provide:

- High resolution
- High refresh rate
- Good brightness
- Low persistence
- Wide field of view

Why resolution matters

Suppose the display resolution is low.

When the image is enlarged close to your eyes, individual pixels can become visible.

This can produce the screen-door effect, where the user feels as if they are looking through a mesh.

Higher resolution reduces this effect.

2. Optical system

The display is very close to the user's eyes.

Normally, your eyes cannot comfortably focus on a display positioned only a few centimetres away.

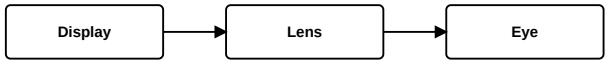
Therefore, the HMD uses lenses.

The optical system:

- Magnifies the display
- Changes the apparent focal distance
- Helps create a wider field of view
- Directs the image toward the user's eyes

A simplified arrangement is:

DIAGRAM



Simplified optical arrangement

Each eye generally receives its own image.

3. Stereo vision

This is extremely important for VR.

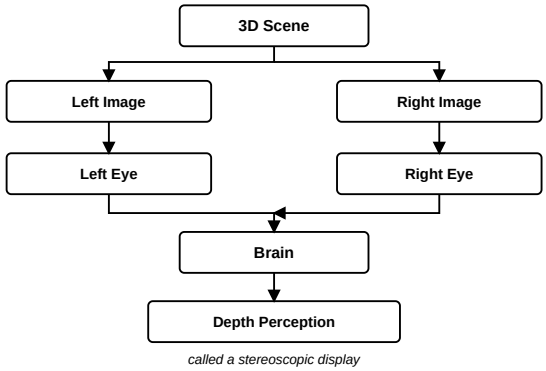
Humans have two eyes separated by a small distance.

Each eye receives a slightly different view.

The brain combines these views to perceive depth.

VR uses the same principle.

DIAGRAM



called a stereoscopic display

Stereoscopic display

This is called stereoscopic display.

4. Tracking system

An HMD needs to know how the user's head is moving.

For example:

You turn your head to the right.

The virtual scene should also change to show the right side.

The headset therefore uses sensors such as:

- Accelerometer
- Gyroscope
- Magnetometer
- Cameras
- Infrared sensors

These determine the user's:

- Position
- Orientation
- Movement

This is called head tracking.

5. Audio system

Modern VR is not only visual.

The headset may contain:

- Speakers
- Headphones
- Spatial audio systems

The system can make sounds appear to come from specific directions.

For example:

```
Virtual car
↓
Sound generated from right side
↓
User hears sound from right
```

This is called spatial or 3D audio.

It greatly improves immersion.

6. Headset enclosure

The physical enclosure:

- Holds the components together
- Blocks unwanted light
- Keeps the display aligned with the user's eyes
- Provides mechanical support

For VR, blocking external light is particularly useful because it prevents the real environment from interfering with the virtual experience.

7. Processing system

The headset needs processing hardware to handle:

- Tracking
- Graphics
- Sensor data
- Display output
- Communication

Modern standalone VR headsets can perform much of this processing internally.

A PC-based headset may receive rendered images from a powerful computer.

8. Adjustment mechanisms

Different users have different eye positions and facial shapes.

Therefore, HMDs may provide:

- IPD adjustment
- Lens adjustment
- Head strap adjustment
- Eye relief adjustment

IPD

Interpupillary Distance (IPD) is the distance between the centers of a person's pupils.

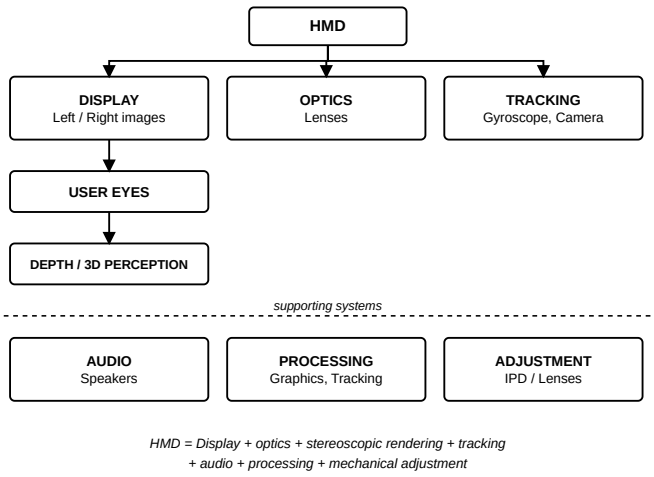
The distance between the two virtual display views should correspond reasonably well to the user's eyes.

Complete HMD architecture

DIAGRAM REQUIRED — VERY IMPORTANT

Learn this:

DIAGRAM



Complete HMD architecture

What you need to remember

An HMD is not simply "a screen attached to your head."

It is an integrated system containing:

Display + optics + stereoscopic rendering + tracking + audio + processing + mechanical adjustment.

3. Fully Immersive Displays

Now let's understand the word immersive.

Imagine playing a normal game on a monitor.

You can see the game, but you still clearly know:

"I am sitting in front of my monitor."

In a fully immersive VR system, the goal is to make the virtual environment occupy most or all of your visual experience.

You should feel:

"I am inside this environment."

That is the basic idea of a fully immersive display.

Definition

A fully immersive display is a display system designed to surround the user's perceptual field with computer-generated information, creating a strong sense of presence within a virtual environment.

It generally involves:

- Wide field of view
- Stereoscopic vision
- Head tracking
- High-resolution displays
- Low latency
- Spatial audio
- Sometimes haptic feedback

Why is immersion important?

Suppose you are in a VR roller coaster.

If you can clearly see the real room around the headset, the illusion becomes weak.

But if your vision is completely occupied by the virtual roller coaster:

```
USER
↓
VIRTUAL WORLD
TRACK
```

your brain receives much stronger virtual-environment information.

Important characteristics of fully immersive displays

1. Wide Field of View

Field of View (FOV) is the extent of the visual scene visible to the user.

A narrow FOV feels like looking through a small window.

A wider FOV gives a more natural visual experience.

2. Stereoscopic display

Separate images are provided to the two eyes.

This produces depth perception.

```
Left eye  → Left image
Right eye → Right image
        ↓
Depth perception
```

3. Head tracking

The system tracks head movement.

If the user turns:

```
User turns head
↓
Tracking
↓
Virtual camera changes
↓
New scene rendered
```

Without this, the environment would feel like a flat picture attached to your face.

4. Low latency

Latency is the delay between the user's movement and the corresponding change in the virtual environment.

For example:

```
Head movement
↓
Tracking
↓
Rendering
↓
Display
```

If this takes too long, the user can experience:

- Discomfort
- Motion sickness
- Loss of immersion

Therefore, low latency is crucial.

5. High refresh rate

The display needs to update frequently.

Higher refresh rates generally make motion appear smoother.

6. Spatial audio

Sound should correspond to the user's position and orientation.

For example, if a virtual helicopter is behind you:

```
Helicopter
↓
Sound
↓
USER
```

the system should make the sound appear to originate from behind.

Types of immersive display approaches

Head-Mounted Display

The user wears a headset.

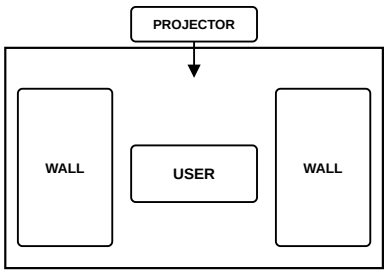
This is the most common modern VR approach.

CAVE

A CAVE (Cave Automatic Virtual Environment) is an immersive room-like environment where images are projected onto multiple walls and sometimes the floor.

For example:

DIAGRAM



images projected onto multiple walls and sometimes the floor

CAVE (Cave Automatic Virtual Environment)

The user stands inside the projected environment.

CAVEs are particularly useful in:

- Engineering
- Scientific visualization
- Training
- Simulation

HMD vs CAVE

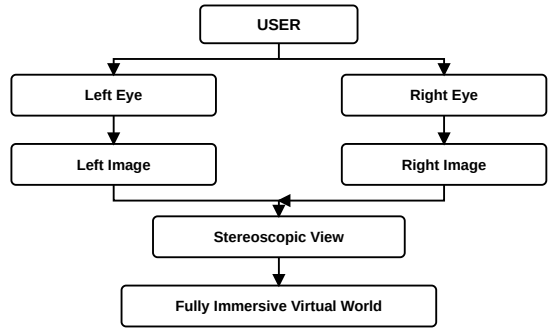
HMD	CAVE
Worn on head	Room-sized system
Portable	Large installation
Personal experience	Multiple people may view environment
Uses headset displays	Uses projectors/screens
Common in consumer VR	Common in research/industrial environments

DIAGRAM REQUIRED

Learn a fully immersive HMD diagram and understand CAVE as an alternative immersive display.

A good exam diagram:

DIAGRAM



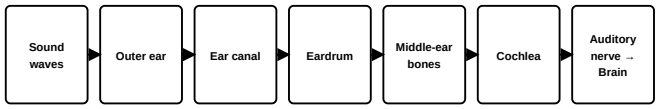
Fully immersive display

4. Mechanics of Hearing

The mechanics of hearing explains how sound waves are collected by the ear, converted into mechanical vibrations and finally transformed into electrical signals interpreted by the brain.

How hearing works

DIAGRAM



How hearing works

Let's understand it.

Outer ear

The outer ear collects sound waves and directs them through the ear canal.

Eardrum

Sound waves cause the eardrum to vibrate.

Middle ear

Three tiny bones amplify and transmit these vibrations.

Cochlea

The cochlea converts mechanical vibrations into neural signals.

Auditory nerve

The signals travel to the brain through the auditory nerve.

Importance in VR

VR systems use spatial audio to create the perception that sounds originate from particular locations.

For example:

```
Virtual car
↓
Sound generated
↓
User hears it from the left
↓
Brain perceives "Car is on my left"
```

This improves the feeling of presence.

3-mark answer

The mechanics of hearing describes how the human ear receives sound waves and converts them into signals perceived by the brain. Sound enters the ear canal and vibrates the eardrum. The middle-ear bones transmit the vibration to the cochlea, where it is converted into neural signals that travel through the auditory nerve to the brain. VR uses this hearing mechanism to produce spatial and 3D audio.

Diagram: Recommended

Learn:

Sound → Ear canal → Eardrum → Middle ear → Cochlea → Auditory nerve → Brain

5. Audio Displays

Now the difference is important:

Mechanics of Hearing = How humans hear.

Audio Display = How the VR/AR system produces sound for the user.

What is an Audio Display?

An audio display is a device or system that presents auditory information to the user.

In VR/AR, audio displays are used to reproduce:

- Speech
- Music
- Environmental sounds
- Warnings
- Spatial sound
- Directional sound

Common devices include:

- Headphones
- Earphones
- Speakers
- Spatial audio systems

Spatial Audio

This is the most important concept.

In real life, your brain can determine approximately where a sound comes from.

For example:

```
Sound source
      \
       \
       USER
```

Your brain receives slightly different sound information at the two ears.

VR systems reproduce these differences to create the perception of sound location.

For example:

```
Virtual helicopter
↓
Spatial audio processing
↓
Different signals to left/right ears
↓
User perceives helicopter as being behind/right
```

This makes the virtual environment more realistic.

Binaural audio

Binaural audio produces separate audio signals for the left and right ears to simulate natural spatial hearing.

The system can manipulate:

- Time difference
- Intensity difference
- Frequency characteristics

between the two ears.

Audio displays in VR

Audio can provide:

Direction

Where is the sound coming from?

Distance

How far away is the sound?

Environment

Does the sound appear to be inside a room, outside, tunnel, etc.?

Events

Audio can indicate an event even when the user is not looking at it.

Example:

You hear footsteps behind you in a VR game.

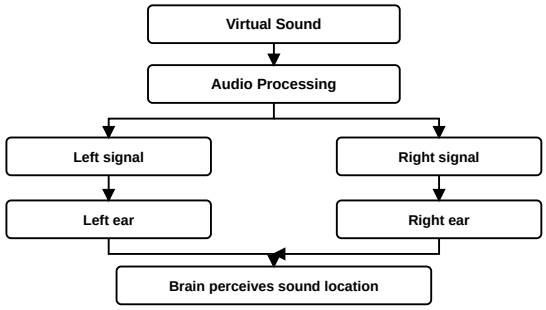
You turn around because the audio tells you where something is happening.

3-mark answer

An audio display is a device or system that presents auditory information to a VR/AR user. Headphones, earphones and speakers are commonly used. VR systems use spatial or binaural audio to create the perception that sounds originate from specific directions and distances. This improves immersion and helps users understand events occurring outside their current visual field.

Diagram: Not essential, but useful

DIAGRAM



Spatial audio processing

6. Mechanics of Feeling

This deals with the sense of touch and how humans perceive physical contact, pressure, vibration and forces.

In VR, this is associated with haptics.

What happens when you touch something?

Suppose you touch a table.

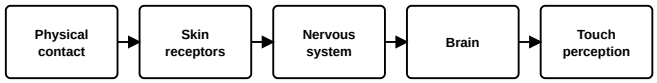
Your skin contains sensory receptors.

When your hand contacts the table, these receptors detect things such as:

- Pressure
- Vibration
- Texture
- Temperature
- Movement

The signals are transmitted through nerves to the brain.

DIAGRAM



How touch works

Types of sensations

Pressure

You can feel how strongly something presses against your skin.

Vibration

You can feel a vibrating object.

Texture

Your sensory system can distinguish rough and smooth surfaces.

Temperature

You can detect hot and cold.

Force

Muscles and sensory receptors help you perceive resistance and physical forces.

Feeling in VR

VR uses haptic devices to create artificial touch sensations.

For example, suppose you touch a virtual button.

```
User touches virtual button
↓
VR detects interaction
↓
Haptic actuator activated
↓
Vibration / pressure
↓
User feels virtual interaction
```

A more advanced force-feedback system can provide resistance.

For example, if a virtual object is solid:

```
Virtual hand → Object
↓
Collision
↓
Force calculation
↓
Force feedback
↓
User's hand
```

The user therefore feels as though the virtual object has physical properties.

Tactile vs force sensation

This connects directly to your 10-mark topic Tactile and Force Feedback Devices.

Tactile feedback → sensations on the skin, such as vibration or pressure.

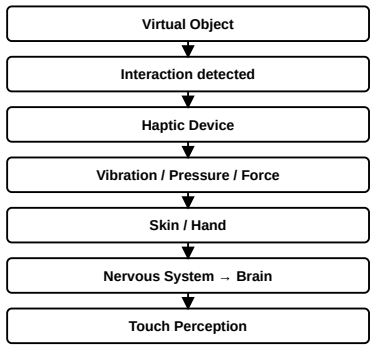
Force feedback → physical resistance or force applied to the user's body.

3-mark answer

The mechanics of feeling describes how the human body detects physical sensations such as pressure, vibration, texture, temperature and force through sensory receptors and transmits them to the brain. In VR, haptic systems reproduce some of these sensations using vibration, pressure or force feedback. This allows users to feel interactions with virtual objects and improves realism and immersion.

Diagram: Recommended

DIAGRAM

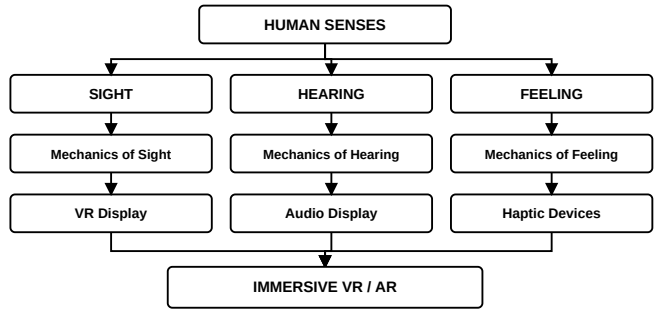


Feeling in VR

The four concepts are connected

This is the easiest way to understand this section:

DIAGRAM



The four concepts are connected

So don't memorize these as four unrelated definitions.

7. Tactile and Force Feedback Devices

Now we move from seeing to feeling.

This is one of the most interesting parts of VR.

Imagine you are holding a virtual ball.

If you can see the ball but your hand feels nothing, the interaction may not feel completely realistic.

VR can therefore provide haptic feedback.

What is haptic feedback?

Haptic feedback is the use of touch-related sensations to communicate information to the user.

It can include:

- Vibration
- Pressure
- Texture sensation
- Resistance
- Force

There are two major concepts in your syllabus:

Tactile feedback

Provides sensations related mainly to touch at the skin surface.

Force feedback

Applies physical forces to the user to simulate resistance or interaction.

Tactile feedback

Suppose you touch a virtual button.

The system may produce a small vibration.

```
Virtual button
↓
User touches
↓
Controller vibrates
↓
User feels vibration
```

This is tactile feedback.

Common examples:

- Game controller vibration
- Haptic motors
- Vibrating gloves
- Tactile displays

Force feedback

Force feedback goes further.

Suppose you are controlling a virtual robotic arm.

When the virtual arm hits a wall, the device can resist your hand movement.

```
User moves hand
↓
Virtual hand hits wall
↓
Collision detected
↓
Force generated
↓
User feels resistance
```

This is force feedback.

Tactile vs Force Feedback

Tactile	Force
Mainly stimulates skin/touch receptors	Applies physical force
Vibration is common	Resistance/pressure is common
Usually simpler	Usually more complex
Example: controller vibration	Example: force-feedback joystick

Tactile and force feedback devices

1. Haptic gloves

Gloves can contain:

- Vibration actuators
- Pressure elements
- Force mechanisms
- Finger tracking sensors

They allow the user to interact with virtual objects using their fingers.

2. Force-feedback joystick

A joystick can resist movement.

For example, when flying a virtual aircraft, the joystick can provide resistance corresponding to simulated forces.

3. Haptic controllers

VR controllers can provide vibrations when:

- You hit an object
- You fire a weapon in a game
- You touch something
- An object is picked up

4. Haptic stylus

A haptic stylus can simulate resistance while interacting with virtual objects.

It is useful in:

- Medical simulation
- CAD
- Scientific visualization

5. Exoskeleton devices

An exoskeleton can apply controlled forces to body parts.

This can be used in:

- Training
- Rehabilitation
- Robotics
- Simulation

Example — Virtual surgery

Imagine a surgeon practicing a virtual operation.

The virtual instrument touches tissue.

The system calculates:

```
Instrument
↓
Virtual tissue collision
↓
Physical interaction calculation
↓
Force feedback
↓
Surgeon's hand
```

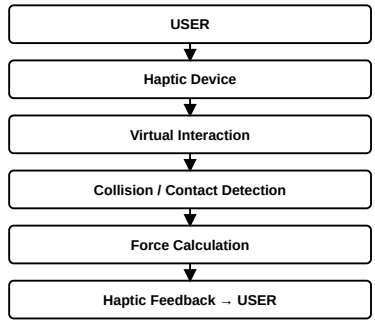
The surgeon can therefore feel resistance.

This makes training more realistic.

DIAGRAM REQUIRED

Learn:

DIAGRAM



Haptic interaction

The important distinction to write in an exam is:

Tactile feedback informs the user through touch sensations such as vibration, whereas force feedback physically applies force or resistance to simulate interaction with virtual objects.

8. Sensors for Tracking, Orientation and Motion

This topic is about one fundamental question:

How does the VR/AR system know what the user is doing?

Suppose you turn your head.

How does the headset know that you turned?

Sensors.

Tracking

Tracking is the process of determining the position and/or movement of an object or user in space.

For VR, the system may need to determine:

- Position
- Orientation
- Velocity
- Movement

Position vs Orientation

This distinction is very important.

Position

Where are you?

For example:

```
X = 2 m
Y = 1 m
Z = 3 m
```

Orientation

Which direction are you facing?

For example:

```
Looking forward
Looking left
Looking right
Looking upward
```

A VR system may track both.

6 Degrees of Freedom

Modern VR tracking commonly considers 6 Degrees of Freedom (6DoF).

They are:

Three translational movements

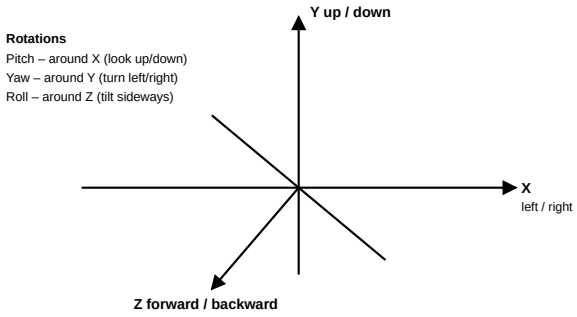
```
X → left/right
Y → up/down
Z → forward/backward
```

Three rotational movements

```
Pitch → looking up/down
Yaw   → turning left/right
Roll  → tilting head sideways
```

Diagram:

DIAGRAM



6 Degrees of Freedom

And rotations:

```
Pitch → rotate around X
Yaw   → rotate around Y
Roll  → rotate around Z
```

This diagram is worth learning.

Important sensors

1. Accelerometer

An accelerometer measures linear acceleration.

It helps detect movement and changes in motion.

For example:

```
Head moves forward
↓
Acceleration changes
↓
Accelerometer detects it
```

2. Gyroscope

A gyroscope measures angular velocity/rotation.

It is extremely important for detecting head rotation.

For example:

```
Head turns right
↓
Gyroscope detects rotation
↓
System updates orientation
```

3. Magnetometer

A magnetometer detects the direction of the Earth's magnetic field.

It can be used as a reference for orientation, particularly heading.

However, nearby metal and electromagnetic interference can affect it.

4. Cameras / Optical Sensors

Cameras can observe the environment and/or tracking markers.

Computer vision algorithms can estimate:

- Position
- Movement
- Orientation

Modern inside-out tracking systems use cameras mounted on the headset to observe the surroundings.

5. Infrared sensors

Infrared systems can be used to detect tracking markers or controller positions.

They are common in optical tracking systems.

6. IMU

An Inertial Measurement Unit (IMU) generally combines:

- Accelerometer
- Gyroscope

and may include a magnetometer.

It provides rapid motion information.

Sensor fusion

No single sensor is perfect.

For example:

- Accelerometers can drift when integrated over time.
- Gyroscopes can accumulate orientation drift.
- Magnetometers can be disturbed by magnetic fields.
- Cameras can have difficulty in poor visual conditions.

Therefore, VR systems often combine sensor information.

This is called sensor fusion.

For example:

```
Accelerometer
+
Gyroscope
+
Camera
↓
Sensor Fusion
↓
Accurate Tracking
↓
Virtual Camera Update
```

This is an important concept.

Inside-out vs Outside-in tracking

Inside-out tracking

Sensors/cameras are mounted on the headset itself.

They observe the environment and determine the headset's movement.

```
Headset Cameras
↓
Observe Environment
↓
Calculate Movement
```

Modern standalone VR headsets commonly use this approach.

Outside-in tracking

External sensors/cameras observe the headset or controllers.

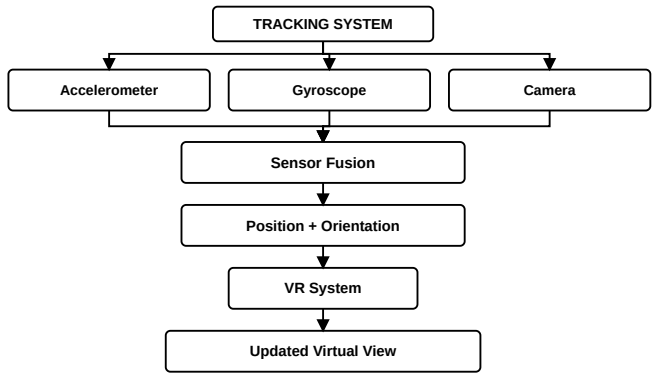
```
External Cameras
↓
Observe Headset
↓
Calculate Position
```

This approach was common in earlier room-scale VR systems.

DIAGRAM REQUIRED — VERY IMPORTANT

Learn this:

DIAGRAM



Tracking system with sensor fusion

9. Devices to Enable Navigation and Interaction

Now we know:

- How the user sees the virtual world
- How the user hears it
- How the user feels it
- How the system tracks movement

But there is one more question:

How does the user control and navigate the virtual world?

That's what this topic covers.

Navigation vs Interaction

These are related but different.

Navigation

Means moving through the virtual environment.

For example:

```
Move forward
Move backward
Turn
Teleport
Walk
Fly
```

Interaction

Means manipulating or communicating with virtual objects.

For example:

```
Pick up object
Press button
Rotate object
Open door
Move object
```

Devices used for navigation and interaction

1. Handheld controllers

VR controllers contain:

- Buttons
- Triggers
- Joysticks
- Motion sensors

They allow the user to:

- Point
- Grab
- Select
- Move
- Rotate

Example:

```
Controller
↓
Point at virtual object
↓
Press trigger
↓
Object selected
```

2. Joystick

A joystick can be used for movement.

For example:

```
Joystick forward
↓
Virtual character moves forward
```

It is commonly used in:

- Flight simulators
- Driving simulators
- VR games

3. Data gloves

Data gloves are wearable input devices that detect hand and finger movements.

They may contain:

- Bend sensors
- Position sensors
- Motion sensors

For example:

```
Finger bends
↓
Sensor detects bend
↓
Computer interprets gesture
↓
Virtual finger bends
```

4. Gesture recognition

Cameras can detect hand gestures without requiring a physical controller.

For example:

```
Open hand → Select
Point → Choose
Pinch → Grab
Wave → Navigate
```

This is especially useful in AR systems.

5. Eye tracking

Eye-tracking systems determine where the user is looking.

This can be used for:

- Selecting objects
- Interface navigation
- Attention analysis
- Foveated rendering

Example:

```
User looks at object
↓
Eye tracker detects gaze
↓
Object highlighted
↓
User selects it
```

6. Voice input

Voice commands can also be used.

For example:

"Open door."

The system recognizes the command and changes the virtual environment.

This is useful when hands are occupied.

7. Treadmills and locomotion devices

Walking naturally in a physical room has limitations because the user eventually reaches a wall.

Specialized locomotion devices can allow users to simulate walking.

Examples include:

- Omnidirectional treadmills
- Walking platforms
- Locomotion interfaces

The user physically walks while the system converts the movement into virtual movement.

8. Touchscreens and mobile devices

AR applications commonly use:

- Touchscreens
- Smartphones
- Tablets

For example:

```
Touch virtual object
↓
Object rotates
```

9. Physical props

A physical object can also be used as an interaction device.

For example, a steering wheel can control a virtual car.

```
Physical steering wheel
↓
Sensor input
↓
VR driving
```

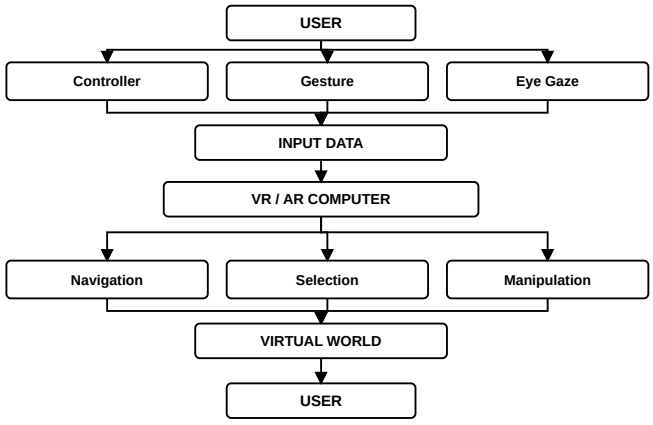
This can make simulation more realistic.

Complete navigation and interaction system

DIAGRAM REQUIRED

Learn this:

DIAGRAM

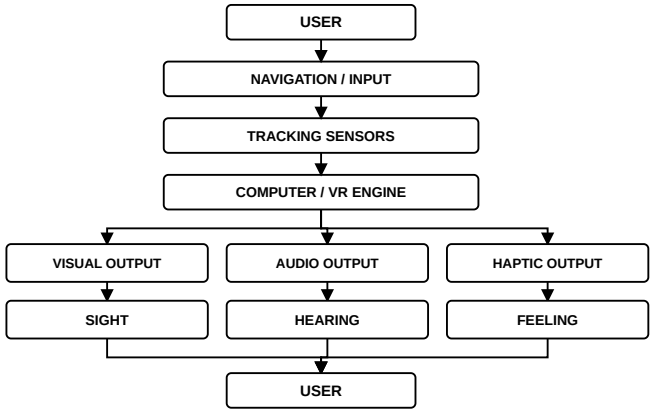


Complete navigation and interaction system

The 5 topics together

Now notice something really important: Module 2 is one complete system.

DIAGRAM



Module 2 as one complete system

MODULE 3 — COMPONENTS OF VIRTUAL REALITY

This module is different from Module 1 and Module 2.

- Module 1: What VR/AR is, how it developed, architecture and applications.
- Module 2: How VR/AR devices provide sight, hearing, touch, tracking and interaction.
- Module 3: How the human brain actually perceives and interprets the information provided by those devices.

So Module 3 is essentially about human perception in VR.

The five 10-mark topics are:

1. Objective and Subjective Reality
2. Perceptual Models and Processes
3. Perceptual Modalities
4. Perception of Space and Time
5. Perceptual Stability, Attention and Action

1. Objective and Subjective Reality

This is the foundation of the whole perception topic.

The first question is:

Is reality exactly what we perceive?

Not necessarily.

There is a difference between what physically exists and what our brain experiences.

Objective Reality

Objective reality refers to the physical world that exists independently of an individual's perception or personal experience.

For example, imagine a red ball is placed on a table.

The ball has physical properties such as:

- Position
- Size
- Shape
- Mass
- Wavelengths of reflected light

These physical properties exist whether or not anyone is looking at the ball.

So:

```
PHYSICAL WORLD
↓
Objects + Physical properties
↓
Objective Reality
```

Subjective Reality

Subjective reality refers to how an individual experiences and interprets the physical world.

The same physical object can produce different experiences in different people.

For example, two people look at the same painting.

One person may think:

"It is beautiful."

Another may think:

"It is boring."

The physical painting is the same.

Their perception and interpretation are different.

Therefore:

```
Objective Reality
↓
Sensory information
↓
Human perception
↓
Brain interpretation
↓
Subjective Reality
```

Why does this happen?

Our brain doesn't directly experience the external world.

Instead:

```
World
↓
Sensory organs
↓
Neural signals
↓
Brain processing
↓
Perception
```

So what we experience is actually the brain's interpretation of sensory information.

This is extremely important for VR.

Example: VR

Suppose you wear a VR headset and see a virtual cliff.

Objectively:

You are actually standing on a flat floor.

But subjectively:

Your brain may experience the virtual cliff as dangerous and high.

You might even experience:

- Fear
- Height sensation
- Increased heart rate
- Hesitation to move forward

Even though there is no actual cliff.

This demonstrates the power of subjective perception.

Why is this important in VR?

VR systems exploit the difference between objective reality and subjective perception.

The computer does not need to physically create a new world.

It only needs to provide sufficiently convincing sensory information so that the brain interprets it as a world.

For example:

```
Computer-generated image
+
Spatial audio
+
Motion tracking
+
Haptic feedback
↓
Human sensory system
↓
Brain interpretation
↓
Perceived virtual reality
```

This is the fundamental idea behind immersive VR.

Optical illusion example

Consider an optical illusion.

The physical image is fixed.

But your brain may perceive:

- movement where there is none
- different shapes
- different sizes
- different colours

This demonstrates that:

Perception is an active process of interpretation, not simply a direct copy of reality.

Objective vs Subjective Reality

Objective Reality	Subjective Reality
Physical world	Individual's experience of the world
Exists independently of perception	Depends on perception and interpretation
Can be measured physically	Experienced psychologically
Same physical object	Experience may differ between people
Example: physical size of a room	Example: room feels large or small

VR example

Imagine a virtual sword.

Objective level

The sword is only:

Computer-generated data + rendered pixels.

Subjective level

The user may experience:

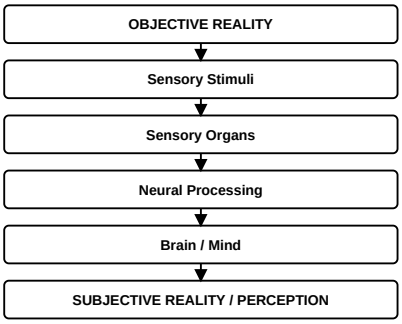
"I am holding a sword."

The subjective experience is what makes VR powerful.

DIAGRAM REQUIRED

Learn this diagram:

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Objective to subjective reality

Core concept

Objective reality is what physically exists; subjective reality is how that reality is perceived and interpreted by an individual.

2. Perceptual Models and Processes

This topic asks:

How does the brain convert sensory information into a meaningful perception?

A VR system produces stimuli.

Your senses receive them.

Your brain processes them.

You perceive an environment.

What is perception?

Perception is the process by which the brain organizes and interprets sensory information to create an understanding or experience of the environment.

For example, your eyes receive light.

Your brain doesn't simply see "millions of pixels."

It interprets those signals as:

"That's a chair."

That interpretation is perception.

Sensation vs Perception

These two terms are important.

Sensation

Detection of physical stimuli by sensory organs.

Example:

Your eyes receive light.

Perception

Interpretation of that sensory information by the brain.

Example:

Your brain identifies the object as a chair.

So:

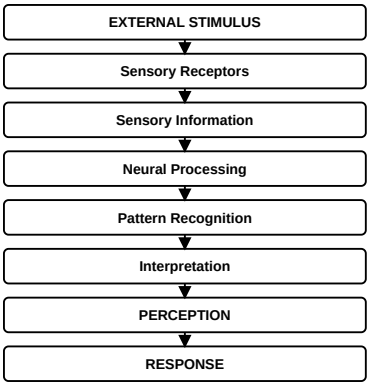
```
Stimulus
↓
Sensation
↓
Neural processing
↓
Perception
```

Perceptual process

A general perceptual process can be represented as:

DIAGRAM REQUIRED

DIAGRAM



The perceptual process

Let's understand each stage.

1. Stimulus

A stimulus is something that activates a sensory system.

Examples:

- Light
- Sound
- Pressure
- Temperature

In VR:

```
VR Display → Light stimulus
Headphones → Sound stimulus
Haptic device → Touch stimulus
```

2. Sensory reception

The sensory organs detect the stimulus.

For example:

```
Light → Eyes
Sound → Ears
Pressure → Skin
```

3. Neural processing

Sensory receptors convert physical information into neural signals.

These signals travel through the nervous system to the brain.

4. Pattern recognition

The brain identifies patterns in the sensory information.

For example, visual information may be interpreted as:

```
Edges
↓
Shape
↓
Object
↓
"Chair"
```

5. Interpretation

The brain combines sensory information with:

- Previous experience
- Context
- Expectations
- Knowledge

to determine what something means.

For example, if you see a familiar object from a distance, your brain can often identify it even though the visual information is incomplete.

Bottom-up and Top-down processing

This is an important concept.

Bottom-up processing

Perception begins with the sensory information itself.

```
Stimulus
↓
Features
↓
Pattern
↓
Object
```

Example:

You see:

- edges
- colour
- shape

and eventually recognize:

"That's a car."

Top-down processing

Existing knowledge and expectations influence perception.

For example, if someone tells you:

"There is a dog behind the door."

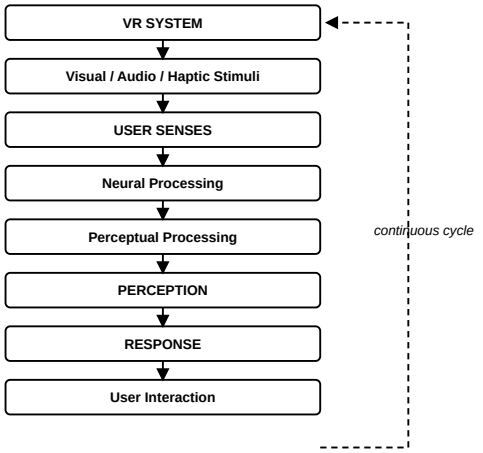
you may interpret an unclear sound as a dog.

Your previous knowledge influences your perception.

Perceptual model in VR

VR provides sensory information that the brain interprets.

DIAGRAM



Perceptual model in VR

This creates a continuous interaction cycle.

Example

Suppose you're playing a VR tennis game.

Stimulus

Virtual tennis ball is rendered.

Sensation

Your eyes detect the moving ball.

Processing

Your brain determines:

- direction
- speed
- distance

Perception

You perceive:

"The ball is coming toward me."

Action

You move the controller.

System response

The virtual racket hits the ball.

This demonstrates that perception and action are closely connected.

Core concept

Perceptual models describe how sensory information is received, processed, interpreted and converted into meaningful perception and action.

3. Perceptual Modalities

This topic is about the different sensory channels through which humans perceive the environment.

A perceptual modality is essentially a particular way in which information is sensed and perceived.

Major perceptual modalities

For VR, the most important ones are:

1. Visual
2. Auditory
3. Haptic/tactile
4. Vestibular
5. Proprioceptive
6. Olfactory
7. Gustatory

You don't necessarily need all of these in every VR system, but understanding them is important.

1. Visual modality

This is perception through the eyes.

It provides information about:

- Shape
- Colour
- Size
- Distance
- Motion
- Depth
- Spatial arrangement

VR primarily relies heavily on visual perception.

```
VR Display
↓
Eyes
↓
Brain
↓
Visual perception
```

2. Auditory modality

This is perception through hearing.

It provides:

- Sound
- Speech
- Direction
- Distance
- Environmental information

VR uses spatial audio to make sounds appear to originate from particular locations.

3. Haptic modality

This involves touch and force sensations.

It can provide:

- Pressure
- Vibration
- Texture
- Force
- Contact

Example:

A VR controller vibrates when your virtual hand touches an object.

4. Vestibular modality

This is related to the inner ear and helps the brain understand:

- Head movement
- Balance
- Acceleration
- Orientation

The vestibular system is extremely important in VR because it interacts with visual information.

For example:

You physically turn your head.

Your vestibular system detects the movement.

Your eyes see the virtual environment moving.

The two signals work together.

5. Proprioceptive modality

Proprioception is the body's ability to sense the position and movement of its own body parts.

For example, close your eyes and raise your hand.

You generally know where your hand is without looking at it.

That's proprioception.

In VR, it helps you understand where:

- Hands
- Arms
- Legs
- Body

are located.

6. Olfactory modality

This is the sense of smell.

Some experimental VR systems can produce smells to increase immersion.

For example:

A virtual forest could be accompanied by an artificial forest-like smell.

7. Gustatory modality

This relates to taste.

It is much less common in VR, but researchers have explored ways to simulate or augment taste experiences.

Multisensory perception

The most powerful VR experiences combine multiple modalities.

For example:

```
Visual → See the fire
Auditory → Hear the fire
Haptic → Feel heat/vibration
Olfactory → Smell smoke
         ↓
  Brain combines
         ↓
Stronger perception
```

This is called multisensory integration.

Sensory conflict

Here's a very important VR concept.

Suppose your eyes tell your brain:

"I am moving forward."

But your body and vestibular system say:

"I am standing still."

This is a sensory conflict.

For example:

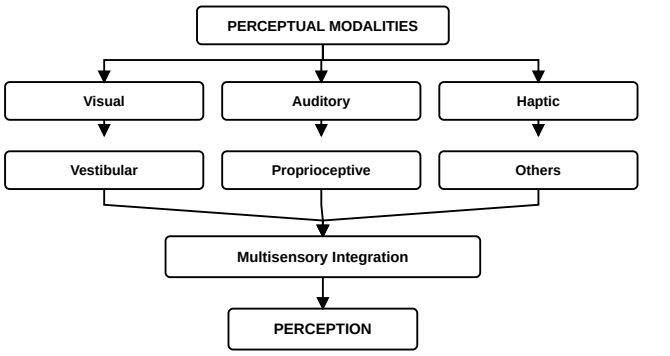
```
Visual system → MOVING
Vestibular → NOT MOVING
        ↓
Sensory conflict
        ↓
Possible discomfort
```

This is one reason some people experience motion sickness in VR.

DIAGRAM REQUIRED

Learn:

DIAGRAM



Perceptual modalities

4. Perception of Space and Time

This topic deals with how humans perceive:

- Where things are
- How far they are
- How they move
- When events happen
- How long events last

So it has two major parts:

Spatial perception + Temporal perception

Part A — Perception of Space

The brain needs to determine the 3D arrangement of the environment.

For example:

```
Tree
 \
  \
  USER
```

Your brain needs to determine:

- Is the tree near or far?
- Is it left or right?
- How large is it?
- How high is it?
- Is it moving?

Depth perception

Depth perception allows us to understand distance and 3D structure.

Several visual cues contribute to depth perception.

Binocular disparity

Each eye receives a slightly different image.

```
Left eye  → Image A
Right eye → Image B
       ↓
     Brain
       ↓
Depth perception
```

VR uses this principle by displaying slightly different images to the two eyes.

Convergence

When looking at a nearby object, the eyes rotate inward toward each other.

The brain can use this information to estimate distance.

```
Eye \     / Eye
     \   /
     Object
```

Motion parallax

When you move your head, nearby objects appear to move more rapidly relative to distant objects.

For example, when looking out of a moving train:

```
Nearby trees → move rapidly
Far mountains → move slowly
```

The brain uses this difference as a depth cue.

Perspective

Objects that are farther away appear smaller.

For example:

```
Near object:  ███
Far object:    ██
Very far:      ▌
```

The brain uses perspective to estimate distance.

Part B — Perception of Time

Humans also perceive the timing and duration of events.

We can distinguish:

- Before
- During
- After
- Duration
- Sequence
- Motion

For VR, timing is extremely important.

Temporal resolution

The brain processes changes over time.

Suppose a VR display responds slowly to your head movement.

You move your head:

```
Head movement
↓
Delay
↓
Virtual scene changes
```

The delay can make the experience feel unnatural.

This is called latency.

Motion perception

The brain determines movement by observing changes in visual information over time.

For example:

```
Frame 1 → Object here
Frame 2 → Object here
Frame 3 → Object here
Frame 4 → Object here
```

The brain interprets the changing positions as motion.

This is why VR requires rapid display updates.

Time perception in VR

VR can manipulate the user's perception of:

- Speed
- Duration
- Timing
- Movement

For example, a virtual object moving rapidly can make an environment feel dynamic even though the user is physically stationary.

Spatial and temporal perception together

These two are strongly connected.

Consider a moving virtual ball.

The brain determines:

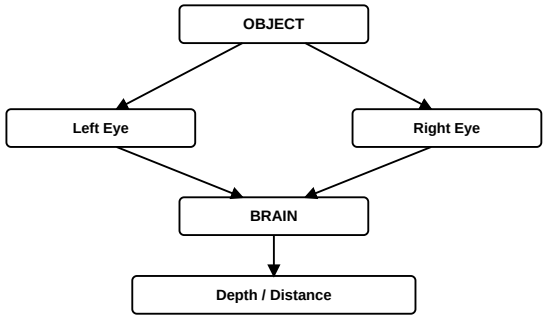
```
WHERE is it?
      +
WHEN will it reach me?
      ↓
Prediction of movement
```

Therefore, VR systems must maintain both spatial accuracy and temporal accuracy.

DIAGRAM REQUIRED

Learn a depth perception diagram:

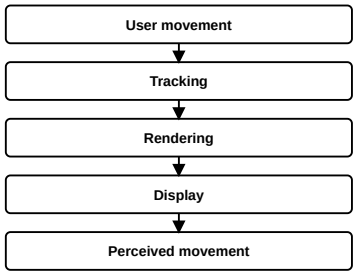
DIAGRAM



Depth perception

And understand the concept of latency:

DIAGRAM



Latency

The smaller the unnecessary delay, the more natural the experience.

5. Perceptual Stability, Attention and Action

This is the final topic, and it combines three related ideas:

1. Perceptual stability
2. Attention
3. Action

The important idea is that perception is not static.

The user is continuously:

perceiving → focusing attention → acting → receiving new information → perceiving again

A. Perceptual Stability

Imagine you move your head while looking at a chair.

The image of the chair moves across your retina.

But you don't normally feel that the chair itself is moving.

Why?

Because the brain compensates for your own movement.

This ability to maintain a relatively stable perception of the environment is called perceptual stability.

Example

You turn your head right:

```
Head moves right
↓
Image on retina changes
↓
Brain knows head moved
↓
Environment perceived as stable
```

The brain combines sensory information with knowledge about self-motion.

Stability in VR

VR needs to maintain perceptual consistency.

Suppose you turn your head right.

The virtual scene must update appropriately.

If it doesn't:

```
Head moves
↓
Virtual scene does not update correctly
↓
Perceptual mismatch
↓
Loss of realism / discomfort
```

Therefore, accurate tracking and low latency are important.

B. Attention

Humans cannot process every piece of sensory information equally.

We selectively focus on some information.

This is called attention.

For example, in a crowded street, you may focus on:

The person talking to you.

while ignoring:

- Other people
- Vehicles
- Background sounds

Attention in VR

VR environments can contain enormous amounts of information.

The system needs to guide the user's attention.

This can be achieved through:

- Lighting
- Colour
- Sound
- Motion
- Object placement
- Animation
- Visual cues

For example:

```
Dark environment
↓
Bright moving object
↓
User's attention
↓
Object becomes focus
```

Selective attention

The user chooses or automatically focuses on particular information.

Example:

In a VR battlefield:

```
Many objects
↓
Enemy movement
↓
Attention shifts
↓
User focuses on enemy
```

Attention and VR design

VR developers use attention deliberately.

For example, if they want you to look at a particular object, they might:

- Make it brighter
- Make it move
- Add sound
- Increase contrast
- Place it in the centre of view

This is important in:

- VR games
- Training
- Education
- Simulation
- Medical applications

C. Action

Perception and action are closely connected.

We don't simply perceive the world.

We use perception to decide what to do.

Example:

```
See virtual ball
↓
Estimate position
↓
Predict movement
↓
Move hand
↓
Hit ball
```

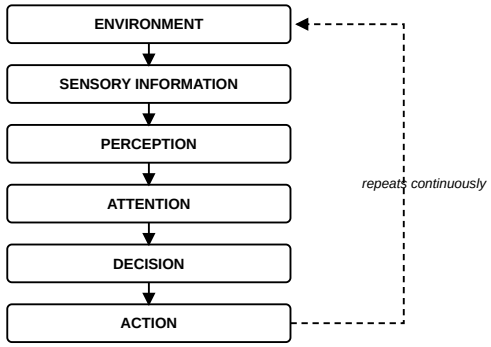
This is called the perception-action loop.

Perception-Action Loop

DIAGRAM REQUIRED — VERY IMPORTANT

Learn this:

DIAGRAM



Perception–Action loop

This loop continuously repeats.

Example: VR tennis

Imagine you're playing VR tennis.

Step 1 — Perception

You see the virtual ball.

Step 2 — Attention

Your attention focuses on the ball.

Step 3 — Interpretation

Your brain estimates:

- Direction
- Speed
- Distance

Step 4 — Action

You move the controller.

Step 5 — Environment changes

The virtual racket hits the ball.

Step 6 — New perception

The ball moves in a new direction.

And the cycle repeats.

```
See ball
↓
Focus
↓
Estimate movement
↓
Move racket
↓
Ball changes direction
↓
See ball again
```

Why is this important in VR?

VR should provide a consistent relationship between:

what the user sees → what the user expects → what the user does → what the system produces.

If these don't match, the user notices the inconsistency.

For example:

You move your head right.

But the virtual scene moves left.

Your brain immediately recognizes something is wrong.

This is a perceptual mismatch.